

INTELLIGENT AML

Ultimate Enterprise Architecture & Execution Blueprint

A Fully Air-Gapped, Neuro-Symbolic Framework for Omni-Channel Financial Fraud
Detection

via Temporal Graph Networks & Offline Agentic AI

Document Scope:

- ✓ The 8-Layer Enterprise / Air-Gapped Stack
- ✓ Hardware Survival Protocol (Core i5 / 16GB)
- ✓ Zero-Copy Data Ingestion Strategy
- ✓ Localized LLM (Ollama) Implementation
- ✓ Complete Contingency Matrix & Roadmap

1. Executive Summary

Legacy anti-money laundering (AML) and compliance infrastructure suffers from a fundamental vulnerability known as **"Relational Blindness."** Traditional tabular ML systems fail to perceive multi-hop topological structures like *Smurfing* and *Peel Chaining*, resulting in unsustainable >95% false-positive rates.

The **Intelligent AML** framework resolves this by deploying a Heterogeneous Temporal Graph Neural Network (HT-GNN) powered by a custom Burst-Aware Edge Attenuation mechanism. This document outlines the transition from a theoretical academic model to an **enterprise-grade, air-gapped B2B SaaS architecture** tailored specifically for banks and MFS providers (e.g., bKash, Dutch-Bangla Bank).

2. The 8-Layer Enterprise Stack (The "Ultimate" Architecture)

This stack strictly enforces a **3-Tier Execution Paradigm**: Python acts as the orchestration layer (Control Plane), while Rust, C++, and CUDA handle all high-performance computations (Execution Plane).

L1: Universal Ingestion Engine

- **Tech:** Polars (Rust) + DuckDB (C++) .
- **Function:** Implements "Lazy Streaming" to convert massive 125GB CSV historical datasets into highly compressed columnar Parquet formats. Queries millions of rows directly from NVMe without loading into RAM (Zero-Copy).

L2: Deep Learning GAN (Data Synthesis)

- **Tech:** Wasserstein GraphGAN (PyTorch) .
- **Function:** Generates structurally valid synthetic money-laundering subgraphs to resolve the severe 97:3 class imbalance between legitimate and illicit transactions.

L3: Core Detection & Federated Privacy

- **Tech:** PyTorch Geometric (PyG) + Flower + Opacus .
- **Function:** Trains the HT-GNN inside isolated bank firewalls. Adds mathematical differential privacy noise so models can learn cross-bank fraud patterns without exposing PII.

L4: Causal Auditing (Explainable AI)

- **Tech:** GNNExplainer + Captum + PyVis .

- **Function:** Eliminates the "Black-Box" liability required by BFIU laws. Masks safe nodes to extract the exact transaction subgraph responsible for triggering an alert.

L5: Air-Gapped Agentic Automation

- **Tech:** `Ollama (Llama 3 8B)` + `CrewAI` + `ReportLab`.
- **Function:** Replaces cloud-based OpenAI dependencies with local LLMs. Autonomous agents parse Layer 4 evidence to draft unalterable, BFIU-compliant Suspicious Activity Reports (SARs) as PDFs in under 60 seconds.

L6: Microservices & Human-in-the-Loop UI

- **Tech:** `FastAPI` + `Streamlit`.
- **Function:** Exposes the AI pipeline as REST endpoints. Provides a dashboard for compliance officers to view 3D PyVis topographies and formally validate alerts.

L7: Identity & Access Management (Security)

- **Tech:** `Keycloak`.
- **Function:** Secures the FastAPI and Streamlit interfaces with OAuth2, JWTs, and Multi-Factor Authentication (MFA). Critical for CISO approval at financial institutions.

L8: Automated Penetration Testing

- **Tech:** `OWASP ZAP`.
- **Function:** Scans the internal API network for vulnerabilities before deployment, automatically generating the security audit PDFs required by banking compliance.

3. The Hardware Survival Protocol

Given the strict local hardware constraints (Core i5 8th Gen, 16GB RAM, H310), traditional execution will result in an immediate Out-Of-Memory (OOM) system freeze. The project utilizes a rigorous two-front strategy:

The 125GB CSV Pre-Compression Fix: Never download raw massive CSVs locally. Use a cloud sandbox (Kaggle/Replit) to run a 3-minute Polars script converting 125GB of CSV text into a ~20GB Parquet file *before* downloading to the local machine.

- **Development (The Local PC):** Handled strictly via lightweight IDEs (Cursor/VS Code). All microservices (FastAPI, Keycloak, Streamlit) are wrapped in docker-compose to enforce hard RAM ceilings (max 8-10GB total).
- **Execution (The Cloud/Kaggle):** Heavy PyTorch matrix multiplication, GraphGAN training, and NeighborLoader batching are executed via cloud GPUs (T4/P100).
- **Production (Bank Deployment):** Packaged via K3s (Lightweight Kubernetes) to run totally offline on a single enterprise banking server (e.g., 32GB RAM, 1x NVIDIA GPU).

4. Omni-Channel Dataset Matrix

Trained across 11 variants to guarantee cross-domain generalization (Crypto, MFS, Fiat).

Phase Group	Dataset Name	Sector Focus	Core Machine Learning Objective
A: Native	DS1/2: Elliptic & Elliptic2	Cryptocurrency	Detecting multi-hop peel chains & mixer usage.
A: Native	DS3: PaySim (PaySim1)	MFS (e.g., bKash)	Mapping high-velocity P2P cash-out smurfing bursts.
B: External	DS4/11: IBM AMLSim	Fiat Banking	Analyzing multi-agent trade-based layering.
B: External	DS5: SynthAML	Institutional	Establishing baseline alert thresholds.
C: Scale	DS6: DGraph-Fin	FinTech Graph	Testing PyG Out-Of-Core NeighborLoader logic.

5. Master Contingency Roadmap (24 Weeks)

Phase	Core Objective	Primary Metric	Fallback / Trigger Condition
1. ETL & Baseline	Zero-copy Parquet Ingestion.	Stream 10M edges in < 5s.	If Polars hits RAM limit -> Downgrade chunk size to 50MB.
2. Generative Auth	GraphGAN minority synthesis.	MMD < 0.05	If unconverged by Week 8 -> Drop GAN, utilize SMOGN.

Phase	Core Objective	Primary Metric	Fallback / Trigger Condition
3. Federated Graph	Opacus DP across institutions.	Epsilon $\leq$ 1.0	If Fed-F1 drops >5% -> Collapse to 2 silos as "limitation".
4. Causal Audits	PyVis Map & GNNExplainer.	Subgraph $\leq$ 15 nodes	If 3D PyVis lags UI -> Downgrade to static Matplotlib.
5. Offline Agents	Ollama + CrewAI SAR drafting.	0% Hallucination	If local Llama3 struggles -> Enforce stricter JSON templates.
6. Enterprise Auth	Keycloak + ZAP Integration.	Secure endpoints.	If K3s fails -> Revert to basic Docker Compose for demo.

End of Master Execution Blueprint v3.0 | Generated for B2B Pitch & Academic Defense